infoblox

NETWORK OBSERVABILITY POWERED BY KENTIK

Kentik Deployment Guide

Prepared for Contoso Manufacturing

Network observability rollout

Customer	Contoso Manufacturing
Project	Network observability rollout
Prepared by	Infoblox SE team
Date	2026-10-29
Kentik region	United States (Ashburn, VA)
Scope	2 data center(s), 4 branch(es), 3 without an agent, AWS, Azure, Synthetic monitoring

This guide describes the Kentik deployment designed for this environment: what gets installed, which permissions and firewall rules are needed, and the steps to go live. Facts are taken from the Kentik Knowledge Base (kb.kentik.com) and each section names its sources.

Infoblox Confidential. Prepared for the named customer only.

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1. Deployment at a glance

Kentik is SaaS. The customer runs a small footprint where the network lives, and everything lands in the customer's Kentik account in the United States (Ashburn, VA) region over encrypted, outbound connections.

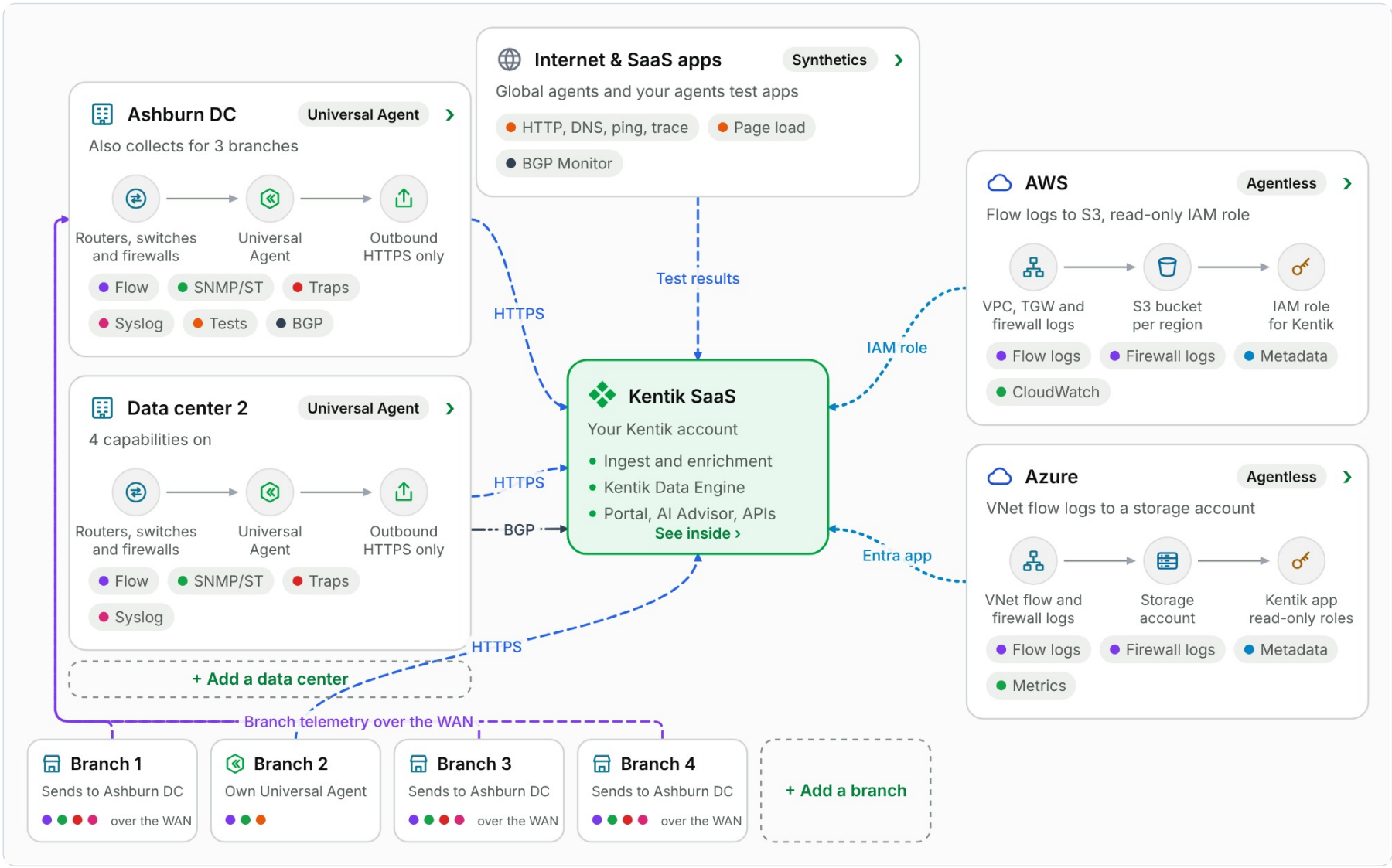
Where	Deployment	What Kentik collects	Footprint
Ashburn DC	Universal Agent on Linux	Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, BGP Proxy	11 vCPU / 20 GB
Data center 2	Universal Agent on Linux	Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server	6 vCPU / 15 GB
Branch 1	No agent	Devices send flow, traps and syslog across the WAN to the Ashburn DC agent	None
Branch 2	Universal Agent on Linux	Flow Proxy: Forwarding, SNMP/ST, Connectivity Test	7 vCPU / 16 GB
Branch 3	No agent	Devices send flow, traps and syslog across the WAN to the Ashburn DC agent	None
Branch 4	No agent	Devices send flow, traps and syslog across the WAN to the Ashburn DC agent	None
Amazon Web Services	Agentless (read-only access)	Topology metadata (required), VPC and Transit Gateway flow logs, Network Firewall logs	No compute
Microsoft Azure	Agentless (read-only access)	Topology metadata (required), VNet flow logs (version 2), Azure Firewall diagnostic logs	No compute

Three patterns cover this design

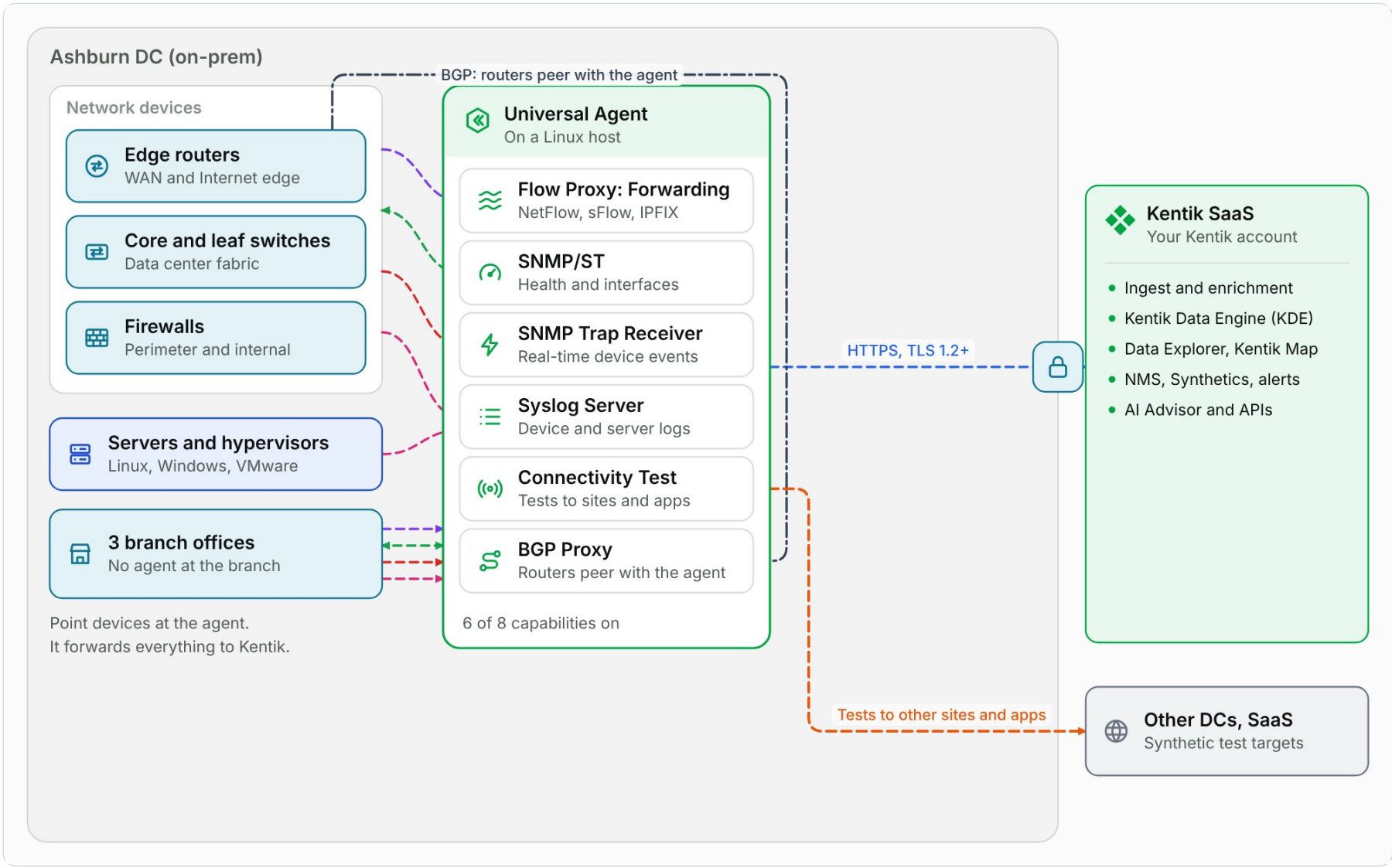
- **On-prem: a Universal Agent per data center.** A single agent with only the capabilities each site needs toggled on in the portal. Devices point at it, and it talks to Kentik over outbound HTTPS. The standalone kproxy agent is not used; the Universal Agent Flow Proxy replaces it.
- **Branches without an agent (3).** Their routers, switches and Wi-Fi send flow, traps and syslog across the WAN to a data center agent, which also polls them. Branches that need local tests or collection get their own agent.
- **Public cloud: agentless.** Flow logs go to storage in the customer's own account, and Kentik gets read-only access to the logs, topology metadata and metrics.
- **Internet and SaaS: synthetic tests.** Kentik's global agents and the customer's own agents test SaaS apps, websites, DNS and paths between sites and clouds.

Source: Kentik KB, Universal Agents; Public Clouds; Synthetics Overview

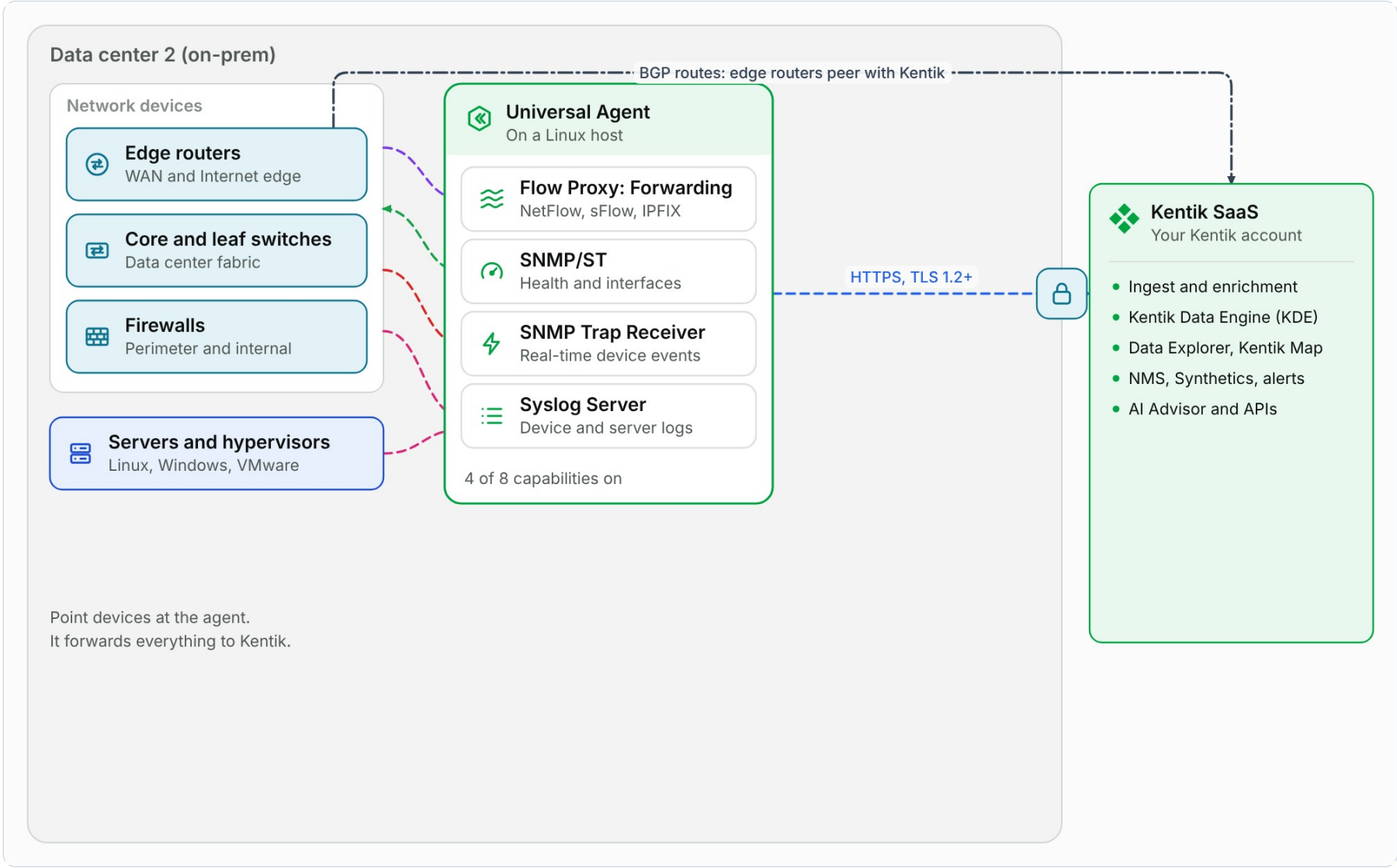
2. Topology


Internet & SaaS apps
Synthetics
Global agents and your agents test apps
HTTP, DNS, ping, trace
Page load
BGP Monitor
Ashburn DC
Universal Agent
Also collects for 3 branches
Routers, switches and firewalls
Universal Agent
Outbound HTTPS only
Flow
SNMP/ST
Traps
Syslog
Tests
BGP
Data center 2
Universal Agent
4 capabilities on
Routers, switches and firewalls
Universal Agent
Outbound HTTPS only
Flow
SNMP/ST
Traps
Syslog
+ Add a data center
HTTPS
HTTPS
BGP
Test results
Kentik SaaS
Your Kentik account
Ingest and enrichment
Kentik Data Engine
Portal, AI Advisor, APIs
See inside ›
IAM role
Entra app
HTTPS
AWS
Agentless
Flow logs to S3, read-only IAM role
VPC, TGW and firewall logs
S3 bucket per region
IAM role for Kentik
Flow logs
Firewall logs
Metadata
CloudWatch
Azure
Agentless
VNet flow logs to a storage account
VNet flow and firewall logs
Storage account
Kentik app read-only roles
Flow logs
Firewall logs
Metadata
Metrics
Branch telemetry over the WAN
Branch 1
Sends to Ashburn DC
over the WAN
Branch 2
Own Universal Agent
Branch 3
Sends to Ashburn DC
over the WAN
Branch 4
Sends to Ashburn DC
over the WAN
+ Add a branch

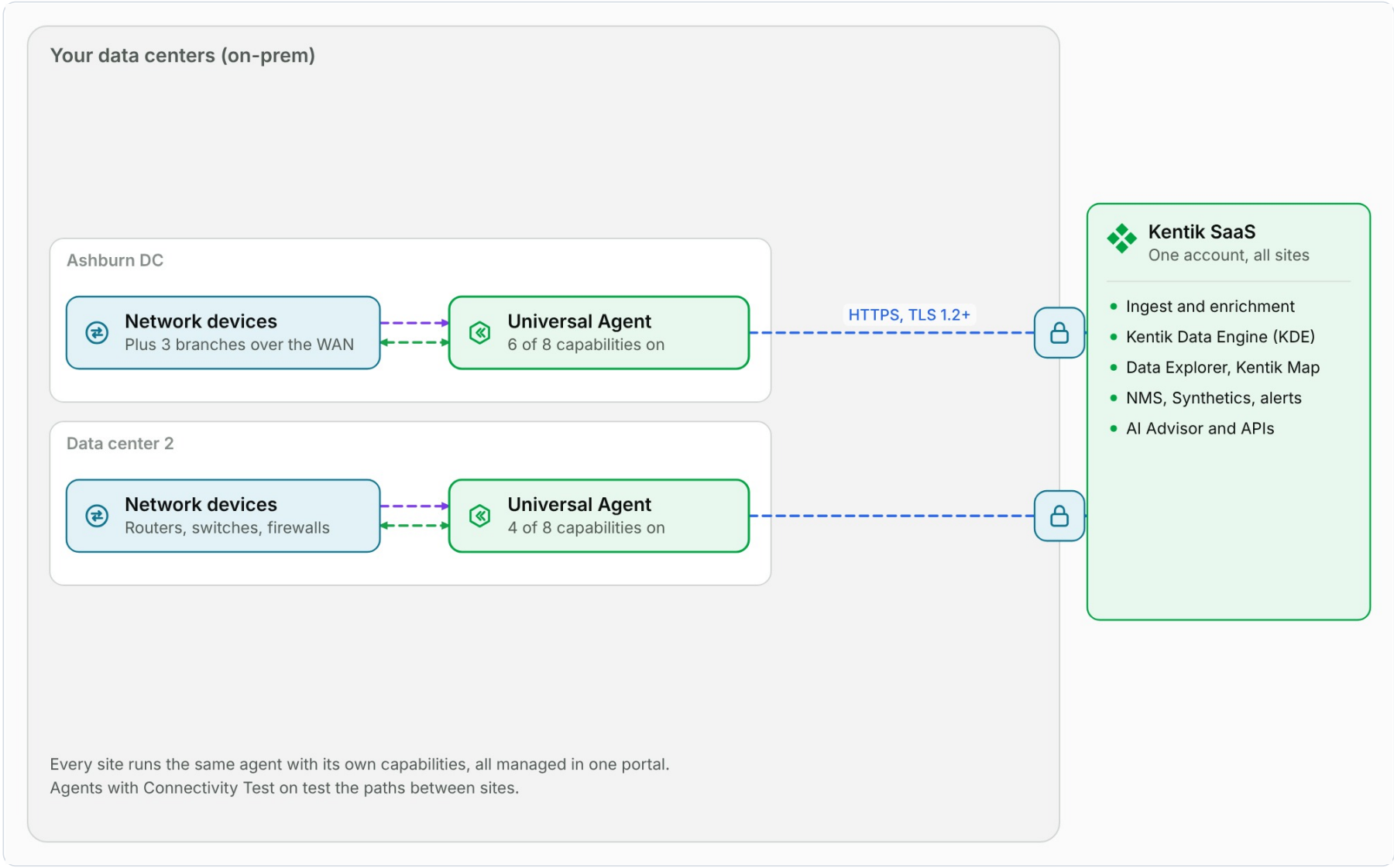
2.1 Ashburn DC



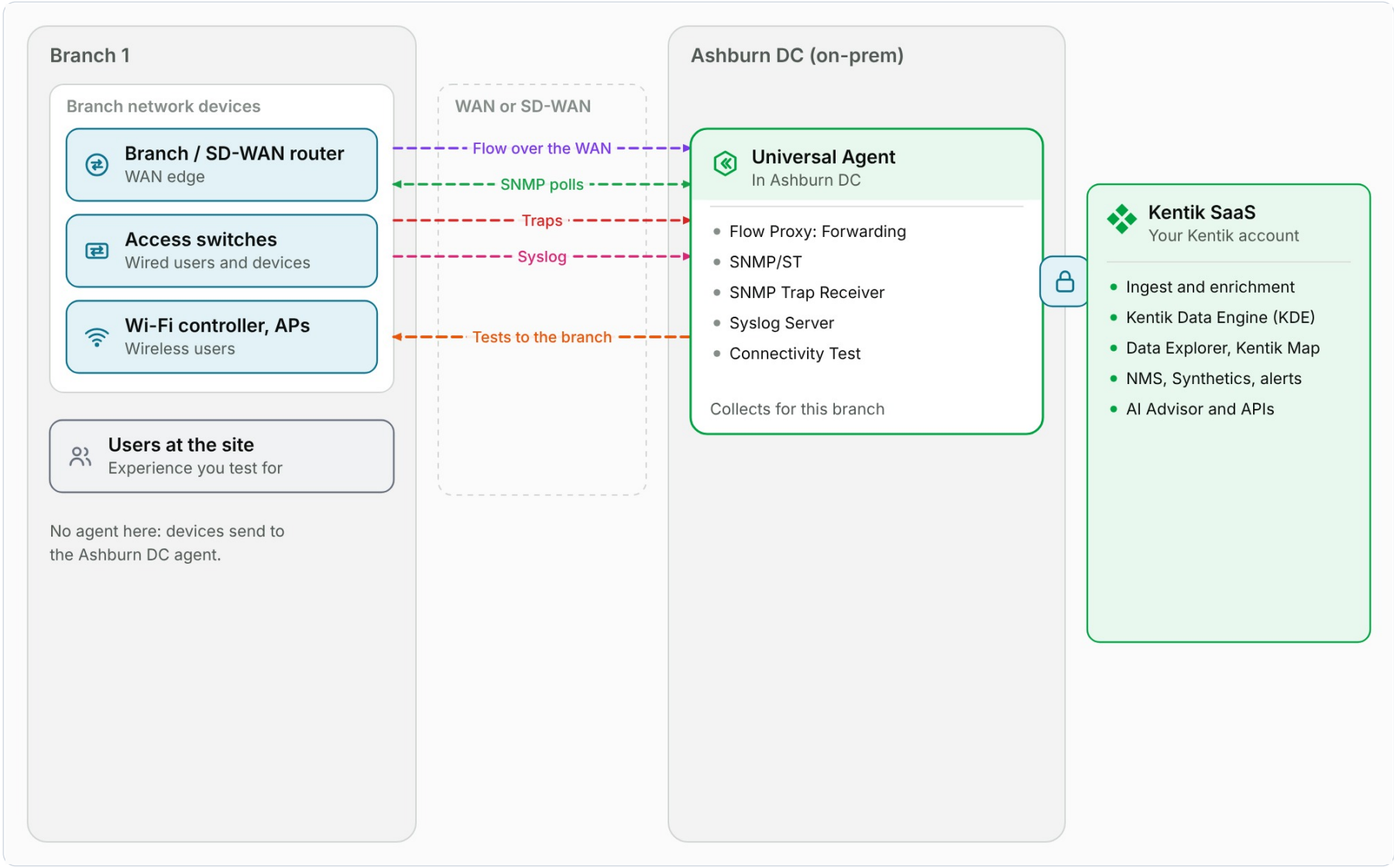
2.2 Data center 2



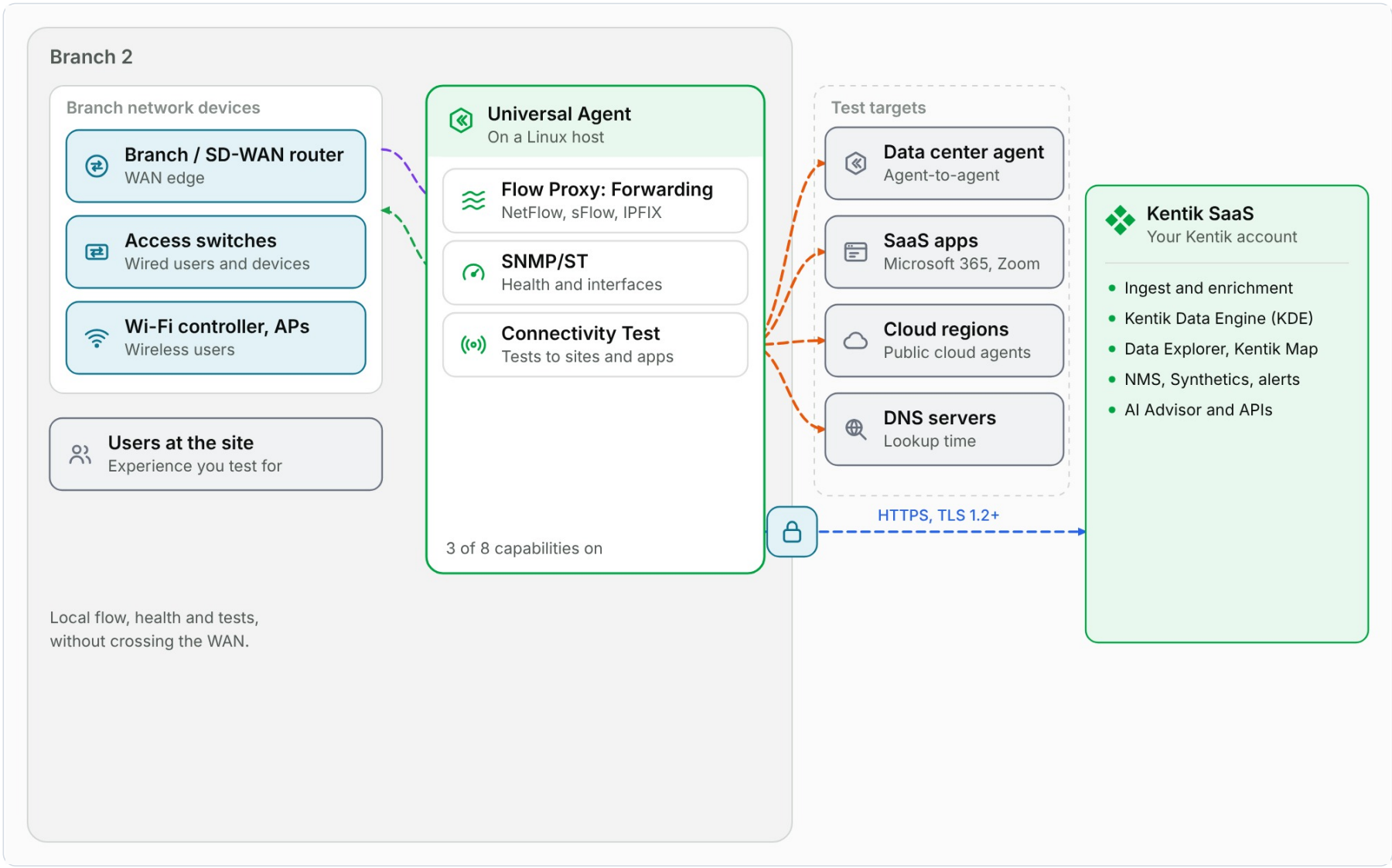
2.3 All data centers



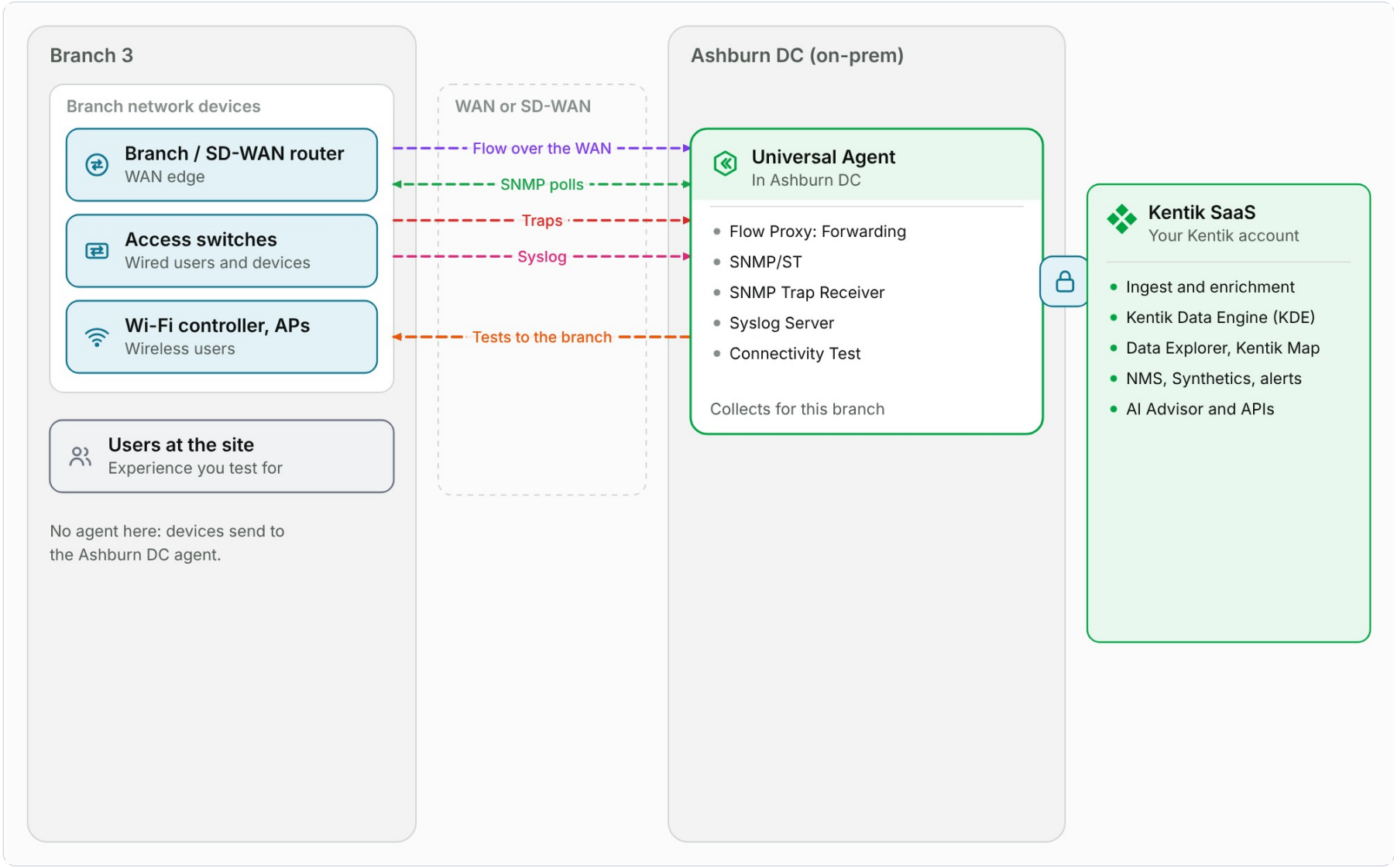
2.4 Branch 1


Branch 1
Branch network devices
Branch / SD-WAN router
WAN edge
Access switches
Wired users and devices
Wi-Fi controller, APs
Wireless users
Users at the site
Experience you test for
No agent here: devices send to the Ashburn DC agent.
WAN or SD-WAN
Flow over the WAN
SNMP polls
Traps
Syslog
Tests to the branch
Ashburn DC (on-prem)
Universal Agent
In Ashburn DC
Flow Proxy: Forwarding
SNMP/ST
SNMP Trap Receiver
Syslog Server
Connectivity Test
Collects for this branch
Kentik SaaS
Your Kentik account
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, alerts
AI Advisor and APIs

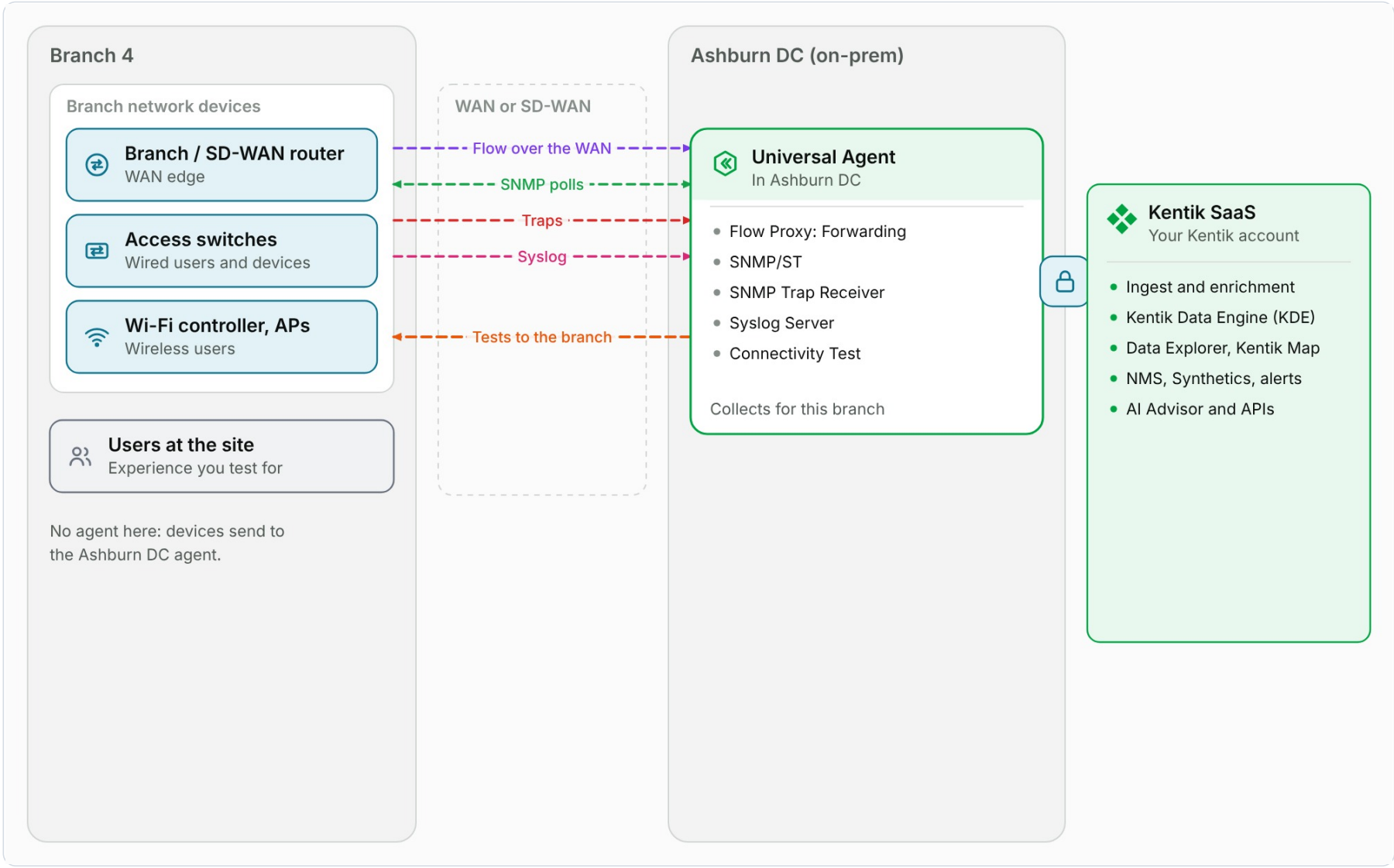
2.5 Branch 2


Branch 2
Branch network devices
Branch / SD-WAN router
WAN edge
Access switches
Wired users and devices
Wi-Fi controller, APs
Wireless users
Users at the site
Experience you test for
Local flow, health and tests,
without crossing the WAN.
Universal Agent
On a Linux host
Flow Proxy: Forwarding
NetFlow, sFlow, IPFIX
SNMP/ST
Health and interfaces
Connectivity Test
Tests to sites and apps
3 of 8 capabilities on
Test targets
Data center agent
Agent-to-agent
SaaS apps
Microsoft 365, Zoom
Cloud regions
Public cloud agents
DNS servers
Lookup time
HTTPS, TLS 1.2+
Kentik SaaS
Your Kentik account
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, alerts
AI Advisor and APIs

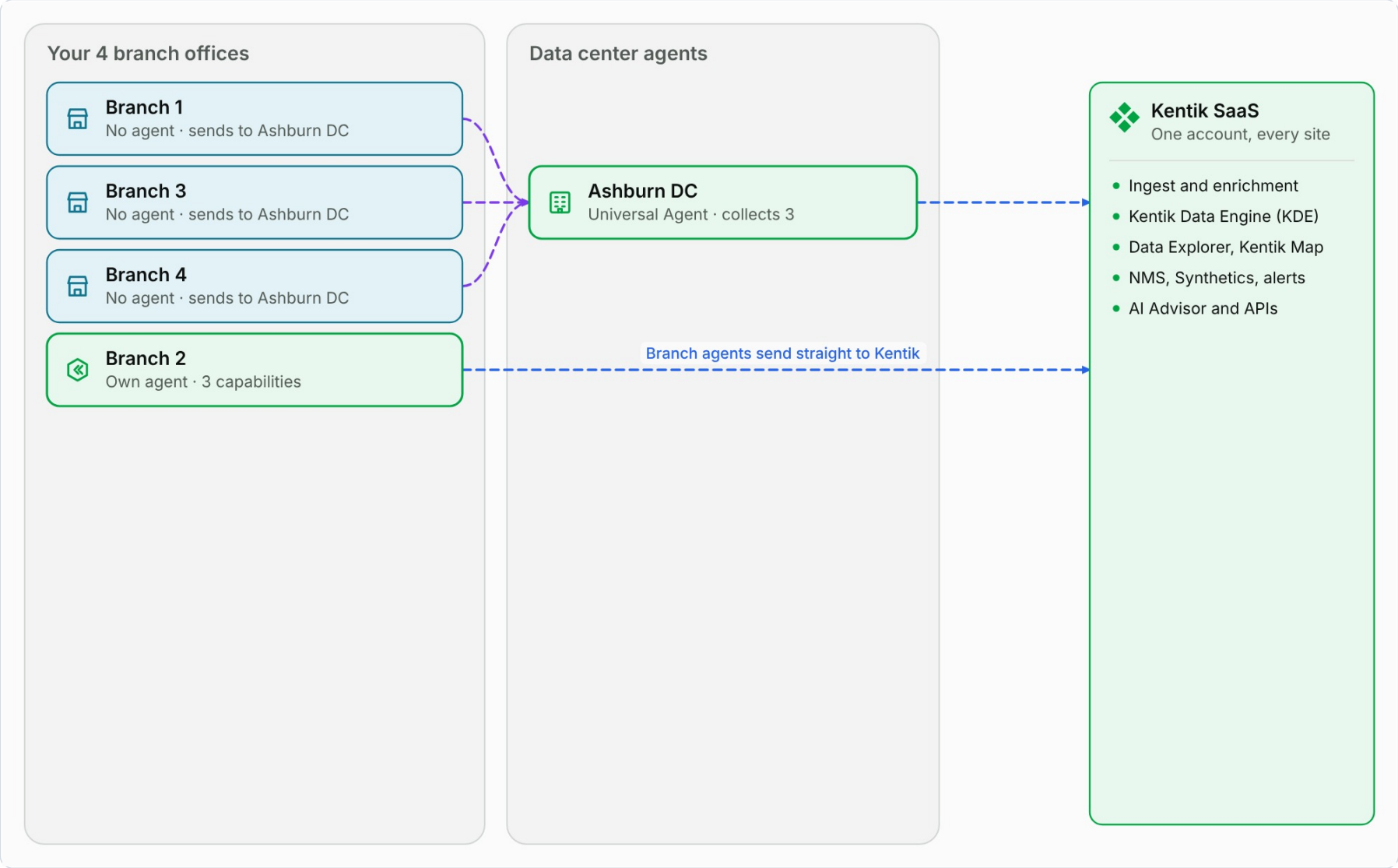
2.6 Branch 3


Branch 3
Branch network devices
Branch / SD-WAN router
WAN edge
Access switches
Wired users and devices
Wi-Fi controller, APs
Wireless users
Users at the site
Experience you test for
No agent here: devices send to the Ashburn DC agent.
WAN or SD-WAN
Flow over the WAN
SNMP polls
Traps
Syslog
Tests to the branch
Ashburn DC (on-prem)
Universal Agent
In Ashburn DC
Flow Proxy: Forwarding
SNMP/ST
SNMP Trap Receiver
Syslog Server
Connectivity Test
Collects for this branch
Kentik SaaS
Your Kentik account
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, alerts
AI Advisor and APIs

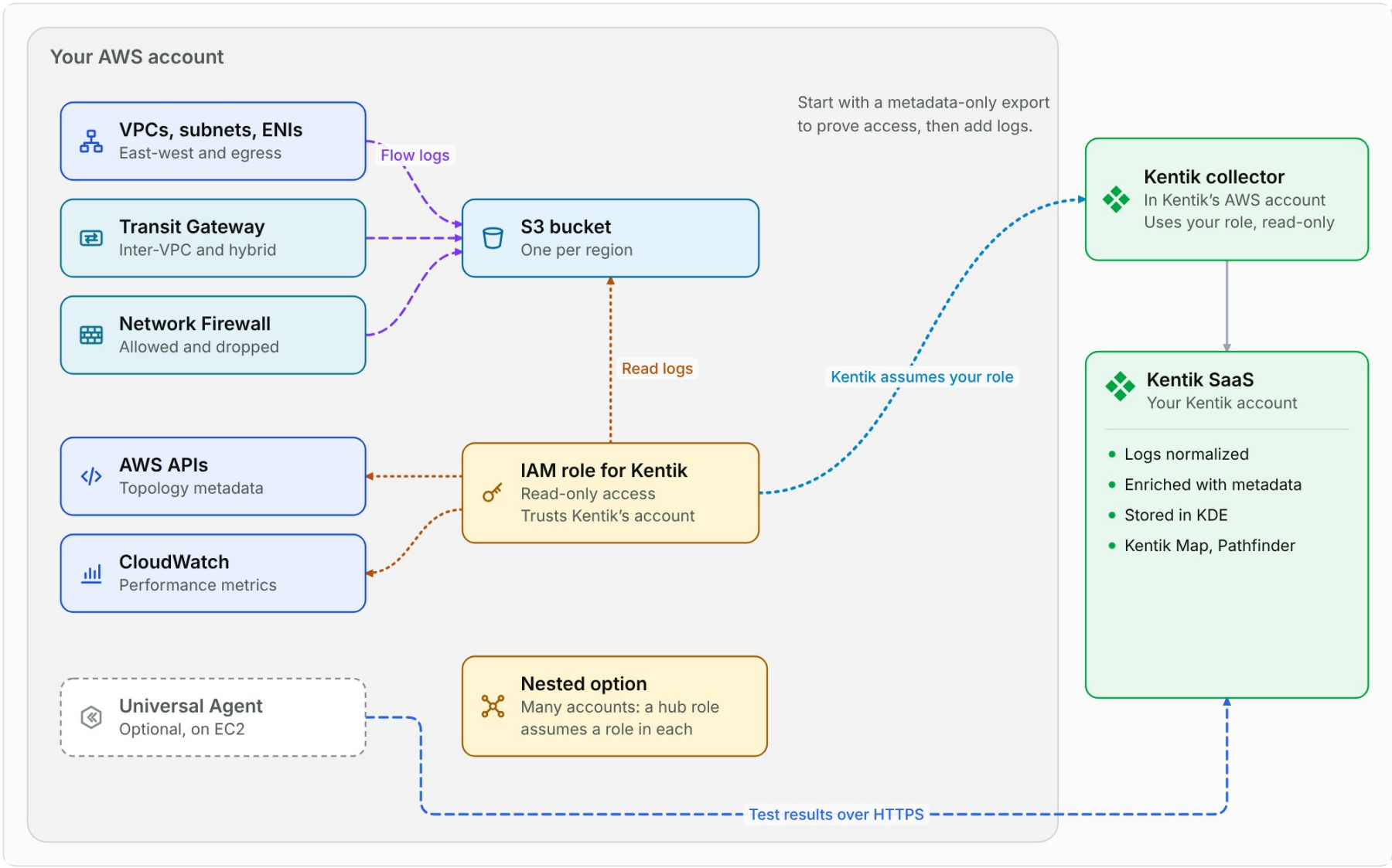
2.7 Branch 4


Branch 4
Branch network devices
Branch / SD-WAN router
WAN edge
Access switches
Wired users and devices
Wi-Fi controller, APs
Wireless users
Users at the site
Experience you test for
No agent here: devices send to the Ashburn DC agent.
WAN or SD-WAN
Flow over the WAN
SNMP polls
Traps
Syslog
Tests to the branch
Ashburn DC (on-prem)
Universal Agent
In Ashburn DC
Flow Proxy: Forwarding
SNMP/ST
SNMP Trap Receiver
Syslog Server
Connectivity Test
Collects for this branch
Kentik SaaS
Your Kentik account
Ingest and enrichment
Kentik Data Engine (KDE)
Data Explorer, Kentik Map
NMS, Synthetics, alerts
AI Advisor and APIs

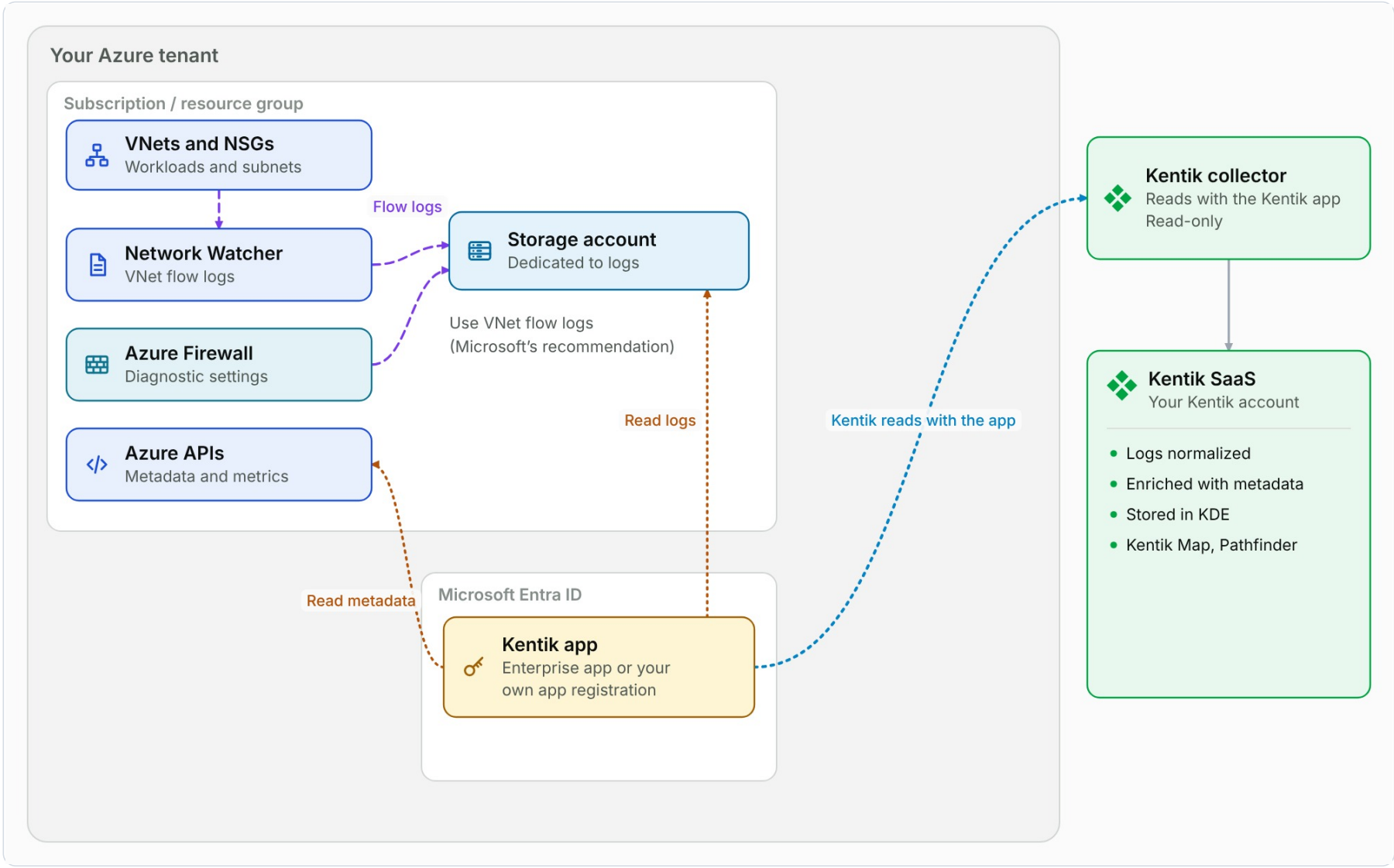
2.8 All branch offices



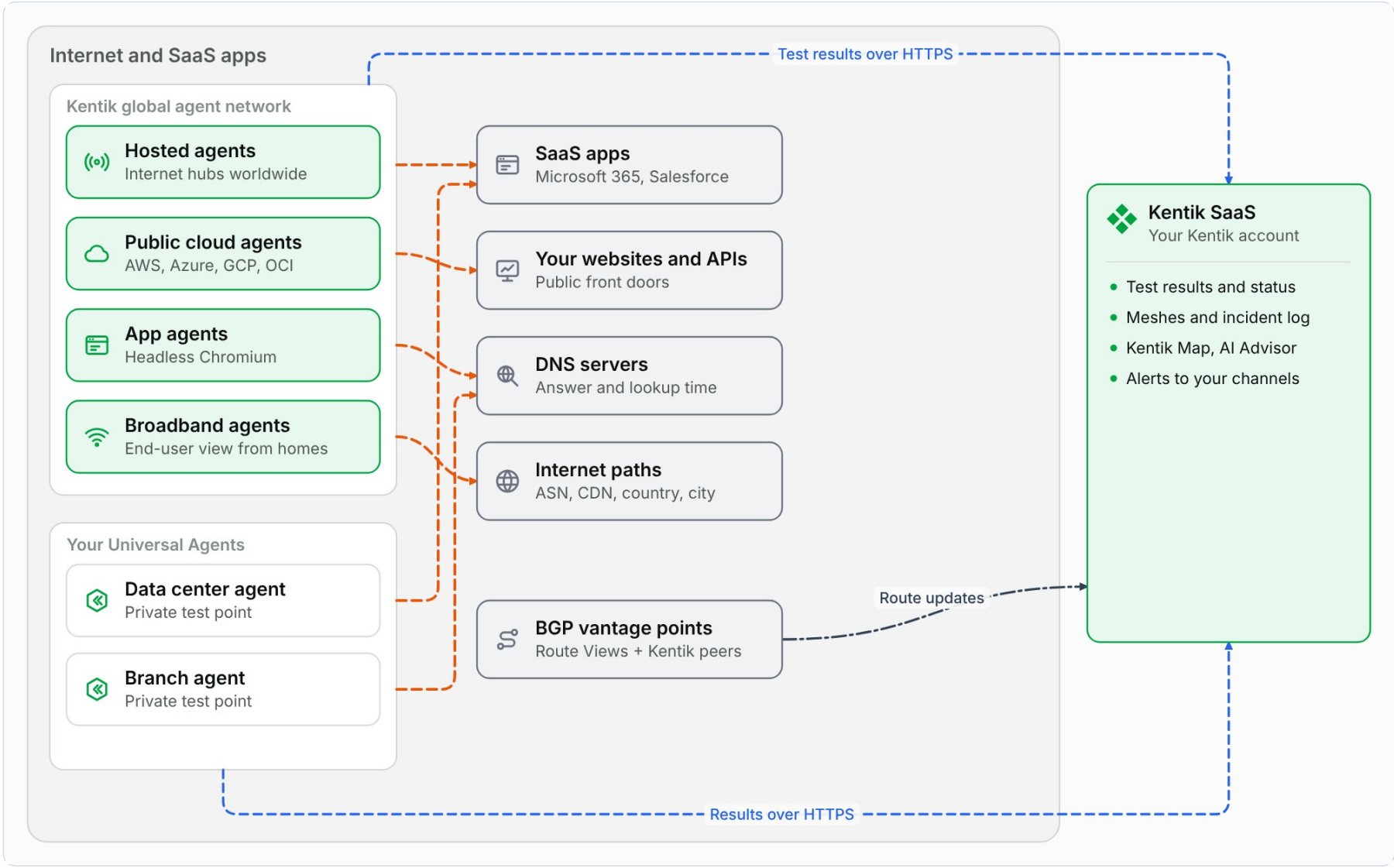
2.9 AWS


Your AWS account
VPCs, subnets, ENIs
East-west and egress
Flow logs
Transit Gateway
Inter-VPC and hybrid
Network Firewall
Allowed and dropped
S3 bucket
One per region
Read logs
AWS APIs
Topology metadata
CloudWatch
Performance metrics
IAM role for Kentik
Read-only access
Trusts Kentik's account
Kentik assumes your role
Start with a metadata-only export to prove access, then add logs.
Kentik collector
In Kentik's AWS account
Uses your role, read-only
Kentik SaaS
Your Kentik account
Logs normalized
Enriched with metadata
Stored in KDE
Kentik Map, Pathfinder
Universal Agent
Optional, on EC2
Nested option
Many accounts: a hub role assumes a role in each
Test results over HTTPS

2.10 Azure


Your Azure tenant
Subscription / resource group
VNets and NSGs
Workloads and subnets
Flow logs
Network Watcher
VNet flow logs
Storage account
Dedicated to logs
Use VNet flow logs
(Microsoft's recommendation)
Azure Firewall
Diagnostic settings
Read logs
Azure APIs
Metadata and metrics
Read metadata
Microsoft Entra ID
Kentik app
Enterprise app or your
own app registration
Kentik reads with the app
Kentik collector
Reads with the Kentik app
Read-only
Kentik SaaS
Your Kentik account
Logs normalized
Enriched with metadata
Stored in KDE
Kentik Map, Pathfinder

2.11 Internet & SaaS apps



3. Roles and permissions

Who on the customer side does what, and the access each task needs. Kentik only ever receives read-only access.

Role	Task	Access needed
Kentik administrator	Deploy and authorize Universal Agents, enable capabilities, add devices, create cloud exports and tests	Kentik admin (or super admin) user in the customer's Kentik account
Server or platform team	Provide 3 agent host(s) and run the install command	root or sudo on a supported Linux host
Network team	Point flow, traps and syslog at the agents (branches without an agent point at their data center agent); create SNMP and SSH credentials; configure BGP peering	Device configuration rights; SNMP v2c/v3 read-only credentials; read-only SSH account
Security / firewall team	Open the rules in section 4	Firewall and proxy change rights
AWS IAM administrator	AWS: enable flow logs and grant Kentik access	IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket
Entra ID Application Administrator and subscription Owner	Azure: enable flow logs and grant Kentik access	Reader on subscriptions or resource groups; Storage Blob Data Reader on the log storage account

Access granted to Kentik

Scope	Access
AWS	IAM role trusting arn:aws:iam::834693425129:root with sts:ExternalId = Kentik Company ID; Describe/List APIs, CloudWatch read, s3:Get* and s3:List* on the log bucket
Azure	Reader on subscriptions or resource groups; Storage Blob Data Reader on the log storage account
Network devices	SNMP read-only (polled from the customer's own agent, never from the Internet)

Source: Kentik KB, UA Deployment; NMS Setup; SSH Command Access for Kentik NMS; Kentik for AWS; Kentik for Azure

4. Network and firewall requirements

All Universal Agent connections are outbound. Nothing needs to reach into the customer network from the Internet.

Outbound allowlist (every agent)

Destination	Port	Notes
flow.kentik.com	TCP 443	TLS 1.2+
grpc.api.kentik.com	TCP 443	TLS 1.2+
api.kentik.com	TCP 443	TLS 1.2+
portal.kentik.com	TCP 443	TLS 1.2+
storage.googleapis.com	TCP 443	TLS 1.2+
whoami.kentiklabs.com	TCP 443	Optional
Kentik flow ingest	UDP 40010	Optional Flow Proxy base port

- SSL-inspecting firewalls and proxies must allow gRPC over port 443.
- UDP 40010 to Kentik is an optional Flow Proxy base port.
- A web proxy works: set https_proxy before running the install (Linux) or pass it with --env (Docker).

Ashburn DC

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Network devices	Universal Agent	UDP 162	SNMP traps
Network devices and servers	Universal Agent	UDP and TCP 514	Syslog
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests
Edge routers	Universal Agent	TCP 179	BGP peering with the agent (BGP Proxy)

Data center 2

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Network devices	Universal Agent	UDP 162	SNMP traps
Network devices and servers	Universal Agent	UDP and TCP 514	Syslog
Edge routers	Kentik ingest peering IPs (shown in device settings)	TCP 179	BGP peering (direct, telemetry only)

Branch 1 · no agent, across the WAN to Ashburn DC

From	To	Port	Purpose
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP 9995	Flow export
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP 162	SNMP traps
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP and TCP 514	Syslog
Ashburn DC Universal Agent	Branch devices	UDP 161, ICMP	SNMP/ST polling across the WAN

Branch 2

From	To	Port	Purpose
Universal Agent	Kentik SaaS (flow.kentik.com, grpc.api.kentik.com, api.kentik.com, portal.kentik.com, storage.googleapis.com)	TCP 443, TLS 1.2+	All telemetry, control and updates (outbound only)
Universal Agent	Kentik SaaS	UDP 40010 (optional)	Flow Proxy base port
Network devices	Universal Agent	UDP 9995	Flow export (NetFlow, sFlow, IPFIX)
Universal Agent	Network devices	UDP 161	SNMP polling (v1, v2c, v3)
Universal Agent	Network devices	ICMP	Reachability
Universal Agent	Network devices	Device gNMI port	Streaming telemetry dial-in (only if used)
Other Universal Agents	Universal Agent (both directions)	TCP 8877 and UDP 9977	Agent-to-agent and mesh tests
Universal Agent	Test targets	ICMP, TCP or UDP	Ping and traceroute

From	To	Port	Purpose
Universal Agent	Test targets	TCP 80 and 443	HTTP(S) and API tests

Branch 3 · no agent, across the WAN to Ashburn DC

From	To	Port	Purpose
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP 9995	Flow export
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP 162	SNMP traps
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP and TCP 514	Syslog
Ashburn DC Universal Agent	Branch devices	UDP 161, ICMP	SNMP/ST polling across the WAN

Branch 4 · no agent, across the WAN to Ashburn DC

From	To	Port	Purpose
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP 9995	Flow export
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP 162	SNMP traps
Branch devices	Ashburn DC Universal Agent (across the WAN)	UDP and TCP 514	Syslog
Ashburn DC Universal Agent	Branch devices	UDP 161, ICMP	SNMP/ST polling across the WAN

Azure storage firewall

Only if the log storage account firewall is restricted: allow 20.69.189.228 and 20.69.185.115 (WestUS2: 51.8.214.254 and 172.171.46.16).

Source: Kentik KB, Universal Agents; UA Deployment; Test Types Reference; Azure Prerequisites & Roles

5. Universal Agent deployment

Install one Universal Agent per site, then switch capabilities on in the portal. The agent updates itself and reports its own health.

Host requirements

- Supported OS: Debian 11, 12, 13; Ubuntu 20.04, 22.04, 24.04; Oracle Linux 9; CentOS Stream 8. CentOS 7 and Enterprise Linux before 8 are not supported.
- Or any Docker host that can run the generated docker run command.
- Size for the sum of the capabilities on the host, plus 20 to 30% headroom for bursty flow, syslog and traps. The plans below use 30%.
- Disk sizing is not published in the KB; confirm with Kentik.

> **Rolling out many agents?** Automated rollout (optional): create a provisioning token with POST https://grpc.api.kentik.com/kagent/v202401/provisioning-tokens (name, maxUsageCount, requiresApproval, config.siteId, optional expiresAt), export K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN on the host, then authorize with PUT .../agents/authorize/<install-id>. A token created with requiresApproval false skips authorization. Revoke the token when all agents have registered.

Source: Kentik KB, Universal Agents; UA Deployment

5.1 Ashburn DC

Data center that also collects for 3 branch(es) without an agent (Branch 1, Branch 3, Branch 4); their load is included in the sizing. Install method: **Linux package**. Capabilities: **Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, BGP Proxy**.

Capability	Planned load	CPU / RAM
Flow Proxy: Forwarding	26,000 flows/s	1 cores / 2 GB
SNMP/ST	260 devices	2 cores / 8 GB
Syslog Server	2,600 events/s	1 cores / 1 GB
Connectivity Test	per agent	2 cores / 2 GB
SNMP Trap Receiver	event driven	Not published
BGP Proxy	per agent	2 cores / 2 GB
Total from KB sizing		8 cores / 15 GB
Recommended host (+30%)		11 vCPU / 20 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.

5. Set Display Name, Description and Site Name (Ashburn DC), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server, Connectivity Test, BGP Proxy) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

5.2 Data center 2

Data center. Install method: **Linux package**. Capabilities: **Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server**.

Capability	Planned load	CPU / RAM
Flow Proxy: Forwarding	20,000 flows/s	1 cores / 2 GB
SNMP/ST	200 devices	2 cores / 8 GB
Syslog Server	2,000 events/s	1 cores / 1 GB
SNMP Trap Receiver	event driven	Not published
Total from KB sizing		4 cores / 11 GB
Recommended host (+30%)		6 vCPU / 15 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Data center 2), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

BGP peering from the edge routers

- In Kentik: device settings > BGP > Peer with device. Kentik shows its ingest peering IPv4 and IPv6 addresses next to the fields.
- Kentik peers as an iBGP route-reflector client using your ASN. 4-byte ASN support is required; an MD5 password is optional.
- Filter the default route toward Kentik so it does not override the destination AS for unmatched flows.

- BGP here is telemetry only; customer routing is unaffected. Peering also enables RTBH and Flowspec mitigation.
- Sites that use the BGP Proxy capability peer with their Universal Agent instead (see BGP Proxy below).

5.3 Branch 2

Branch office. Install method: **Linux package**. Capabilities: **Flow Proxy: Forwarding, SNMP/ST, Connectivity Test**.

Capability	Planned load	CPU / RAM
Flow Proxy: Forwarding	2,000 flows/s	1 cores / 2 GB
SNMP/ST	20 devices	2 cores / 8 GB
Connectivity Test	per agent	2 cores / 2 GB
Total from KB sizing		5 cores / 12 GB
Recommended host (+30%)		7 vCPU / 16 GB RAM

Install and register

1. In the Kentik portal go to Settings > Universal Agents and click + Deploy Agent.
2. Under Installation pick Linux. The portal generates a command that carries K_COMPANY_ID and K_REGISTER_PROVISIONING_TOKEN.
3. Copy the command and run it on the agent host. Portal tokens last one hour, and you can register as many agents as you like in that window.
4. The agent appears under Select an Agent (host name is the system hostname). Click Authorize. An agent left unauthorized shows Not Authorized and does nothing.
5. Set Display Name, Description and Site Name (Branch 2), then click Continue.
6. Toggle on the capabilities for this site (Flow Proxy: Forwarding, SNMP/ST, Connectivity Test) and click Finish.
7. Confirm the agent status is Up. An agent not seen for 30 minutes shows Down. Agents update themselves and their capabilities.

```
export K_COMPANY_ID=<Kentik company ID>
export K_REGISTER_PROVISIONING_TOKEN=<token>
# then run the Linux command copied from Settings > Universal Agents > Deploy Agent
```

Source: Kentik KB, UA Deployment; UA Portal Management; Device Settings

6. Branches without an agent

These branches get no agent. Their devices are configured to use a data center Universal Agent as their collector, across the WAN. Nothing is installed at the branch.

Branch	Sends to	What reaches Kentik
Branch 1	Ashburn DC agent	Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server
Branch 3	Ashburn DC agent	Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server
Branch 4	Ashburn DC agent	Flow Proxy: Forwarding, SNMP/ST, SNMP Trap Receiver, Syslog Server

Steps for each branch

1. On the branch WAN or SD-WAN router (and switches if wanted), export flow to the data center agent's IP on UDP 9995. Set the same sample rate for the device in Kentik.
2. Point SNMP traps at the agent on UDP 162 and syslog on UDP or TCP 514.
3. Allow those flows through WAN firewalls and SD-WAN policy, plus UDP 161, ICMP and TCP 22 from the agent back to the branch devices.
4. In Kentik, add the branch devices with the data center agent as their collection agent, and assign them to the branch Site so they still group by location.
5. Optional: create Server IP or agent-to-agent tests from the data center agent to the branch router to watch the WAN path.

> A branch gets its own agent when it needs tests from where users are, is large or far from a data center, or should keep its telemetry off the WAN.

Source: Kentik KB, Universal Agents; Device Settings; Router Configuration

7. Capability configuration

How to configure each capability used in this design, on the agent and on the devices that feed it.

Flow Proxy: Forwarding

Receives NetFlow, sFlow and IPFIX from routers, switches and firewalls, enriches it and forwards it to Kentik over TLS.

Enabled at: Ashburn DC, Data center 2, Branch 2 · **License:** FlowPak

- Toggle on Flow Proxy: Forwarding on the agent. Settings: Host 0.0.0.0, Port 9995 (UDP), optional sFlow agent address override, and Site.
- On each device, export flow to the agent's IP on UDP 9995. Sample ingress on every interface; never ingress and egress on the same interface.
- The sample rate on the device must match the rate set for that device in Kentik.
- Vendor configuration snippets for flow, SNMP and BGP: github.com/kentik/config-snippets.
- Sanity check after go-live: flow volume should land within about 20% of SNMP interface traffic.

Source: Kentik KB, Universal Agents; Router Configuration; Flow Overview; Migration Scenarios (kproxy to UA)

SNMP/ST

Discovers devices and collects health, interface and hardware metrics by SNMP polling or gNMI streaming telemetry. It also enriches flow with interface names and speeds.

Enabled at: Ashburn DC, Data center 2, Branch 2 · **License:** FlowPak (flow enrichment) or NMS / DevicePak (full monitoring)

- Run the NMS Setup Wizard: pick this agent, enter IP ranges and SNMP v1, v2c or v3 credentials, discover, then add devices.
- Per device, on the SNMP tab choose "Flow Enrichment via Universal Agent" or "Full Monitoring via Universal Agent".
- SNMP v3 options: auth None, MD5 or SHA; privacy None, DES or AES.
- Default polling: 1 minute for status, CPU, memory and interfaces; 5 minutes for everything else.
- Streaming telemetry (optional): enable "Dial-In from Universal Agent" (gNMI over gRPC, 30 second default).

Source: Kentik KB, Universal Agents; NMS Setup; Device Settings

SNMP Trap Receiver

Captures SNMP traps the moment a device sends them, for example an interface or BGP session going down.

Enabled at: Ashburn DC, Data center 2 · **License:** NMS / DevicePak

- Enable SNMP Trap Receiver on the agent.
- Point device trap targets at the agent's IP on UDP 162.
- Traps must come from a device licensed for NMS. Query them in Data Explorer with the metric "SNMP Traps: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

Syslog Server

Collects log messages from devices and servers so they can be searched, alerted on and lined up with traffic.

Enabled at: Ashburn DC, Data center 2 · **License:** NMS / DevicePak

- Enable Syslog Server on the agent.

- Point device and server syslog at the agent's IP on UDP or TCP 514.
- If a syslog forwarder sits in between, it must keep the original device IP. Changing the port currently needs Kentik support.
- Query in Data Explorer with the metric "Syslog: Events".

Source: Kentik KB, Universal Agents; SNMP Traps & Syslog

Connectivity Test

Makes this agent a private test point. It measures latency, loss and jitter to other sites, clouds, SaaS apps and DNS servers, and gives AI Advisor on-demand ping and traceroute.

Enabled at: Ashburn DC, Branch 2 · **License:** Synthetics test credits

- Toggle on Connectivity Test on the agent.
- Open TCP 8877 and UDP 9977 between every pair of agents that will test each other.
- Create tests in Synthetics: agent-to-agent, network mesh, server IP or hostname, DNS server monitor and HTTP(S) or API tests.
- Confirm which continuous test types run on the Universal Agent today with the Kentik account team.

Source: Kentik KB, Universal Agents; Synthetics Overview; Test Types Reference; Test Credits

BGP Proxy

Lets routers peer BGP with the agent instead of with Kentik. The agent relays routes to Kentik over outbound HTTPS, so routers in private networks need no session to the Internet.

Enabled at: Ashburn DC · **License:** FlowPak (BGP enrichment of flow)

- Toggle on BGP Proxy on the agent.
- On each edge router, configure a BGP session to the agent's IP on TCP 179 (in Kentik device settings, BGP tab, pick the agent as the peer).
- Filter the default route toward the agent so it does not override the destination AS for unmatched flows.
- BGP here is telemetry only; routing is unaffected.
- BGP Proxy shows in Installed Capabilities in the portal, while the KB page still lists it as coming soon: confirm availability for the account with the Infoblox account team.

Source: Kentik KB, Universal Agents; Device Settings

8. Public cloud setup

No agents are needed in public clouds. Each cloud writes flow logs to storage in the customer's own account, and Kentik reads them with read-only access. Kentik recommends a metadata-only export first to prove access before turning on high-volume logs.

Amazon Web Services

No agent. Flow logs are written to S3 buckets the customer owns, and Kentik assumes a read-only IAM role protected by an ExternalId.

Collecting: Topology metadata (required), VPC and Transit Gateway flow logs, Network Firewall logs.

Steps

1. Start with a metadata-only export to prove access before turning on high-volume logs.
2. Create one S3 bucket per region for flow logs. Kentik treats each bucket as one device, so hundreds of VPCs into one global bucket risks S3 throttling.
3. Encryption: SSE-S3 is recommended. SSE-KMS needs the KMS statement. SSE-C is not supported.
4. Turn on flow logs for VPCs, Transit Gateways and Network Firewalls: filter All, destination S3, custom format with every v2 to v5 field.
5. Create the IAM role with the trust policy and permissions below.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > AWS Cloud. Select features, enter the Role ARN > Verify Role, Region > Verify Region, bucket name and prefix > Verify S3 Bucket.
7. Pick sampling (Legacy, a rate from 2 to 2000, or Unsampled), name the export, choose the billing plan and Save.
8. Optional: tick "Help me configure my provider via Terraform" for a main.tf that creates the role, bucket and flow logs.

IAM role trust policy

```
{
  "Version": "2012-10-17",
  "Statement": [{
    "Effect": "Allow",
    "Principal": { "AWS": "arn:aws:iam::834693425129:root" },
    "Action": "sts:AssumeRole",
    "Condition": {
      "StringEquals": { "sts:ExternalId": "<Kentik Company ID>" },
      "ArnEquals": { "aws:PrincipalArn": "arn:aws:iam::834693425129:role/eks-ingest-node" }
    }
  }]
}
```

IAM permissions policy

Service	Actions
EC2	ec2:Describe*, ec2:Search*, ec2:GetManagedPrefixListEntries
Direct Connect	directconnect:List*, directconnect:Describe*
ELB	elasticloadbalancing:Describe*
IAM	iam:ListAccountAliases

Service	Actions
Network Firewall	network-firewall:Describe*, network-firewall:List*
Cloud WAN	networkmanager:ListCoreNetworks, GetCoreNetwork, GetCoreNetworkPolicy, ListAttachments, GetNetworkRoutes
CloudWatch	cloudwatch:ListMetrics, GetMetricStatistics, GetMetricData, Describe*
Organizations	organizations:ListAccounts
S3 (log bucket)	s3:Get*, s3:List* on arn:aws:s3:::<bucket> and arn:aws:s3:::<bucket>/*
KMS (SSE-KMS only)	kms:Decrypt, kms:DescribeKey on the bucket key

Export status values: Start, Pending, Error, OK, Disabled or Halt. Licensing: CloudPak, measured in flows per second.

Source: Kentik KB, Kentik for AWS; Metadata Configuration (AWS); Flow/Firewall Log Collection Configuration (AWS); Kentik Export Configuration for AWS; Public Clouds

Microsoft Azure

No agent. Network Watcher writes VNet flow logs to a storage account the customer owns, and Kentik reads it through an app approved in Entra ID.

Collecting: Topology metadata (required), VNet flow logs (version 2), Azure Firewall diagnostic logs.

Steps

1. Create a dedicated storage account for the logs.
2. If the storage account firewall is restricted, allow 20.69.189.228 and 20.69.185.115 (WestUS2 uses 51.8.214.254 and 172.171.46.16).
3. In Network Watcher turn on VNet flow logs (version 2) to that storage account.
4. Send Azure Firewall diagnostic logs to the same storage account with the azfwapplicationrule and azfwnetworkrule categories.
5. Authorize Kentik: an Application Administrator assigns the Kentik enterprise app (Kentik NSG Flow Exporter), then Validate Service Principal in the export wizard.
6. In Kentik: Settings > Public Clouds > Create Cloud Export > Azure Cloud. Pick features, API access method, location and subscription ID, Validate Service Principal, resource group > Verify Access.
7. Enter the storage account name, sampling (2 to 2000 or Unsampled), optional enrichment scope and billing plan, then Save.
8. Optional: the export wizard can generate a PowerShell script that does most of the Azure side (VNet flow logs only).

Roles for the Kentik service principal

Role	Scope	Why
Reader	Subscription or resource group holding the logs	Read resources
Reader	Additional subscriptions or resource groups	Enrichment scope
Storage Blob Data Reader (or Reader and Data Access)	Log storage account	Read exported logs
Custom read-only Kentik role (optional)	Tenant	Full traffic path views

Source: Kentik KB, Kentik for Azure; Azure Prerequisites & Roles; Flow/Firewall Log Collection (Azure); Kentik Export Configuration for Azure; Automated Configuration (PowerShell)

9. Synthetic monitoring plan

Continuous tests measure latency, loss and jitter between sites, clouds and the applications users depend on.

Test points

Agent	Type
Ashburn DC	Private (Universal Agent)
Branch 2	Private (Universal Agent)
3 Kentik global agent(s)	Kentik-hosted: Internet hubs, cloud regions, app agents

Planned tests

- Network mesh between all private agents (latency, loss, jitter between every pair of sites and clouds).
- HTTP(S) test to Microsoft 365 from private and global agents.
- HTTP(S) test to Salesforce from private and global agents.
- BGP Monitor for the customer's prefixes: hijacks, leaks and RPKI status.
- Available test types: Agent-to-Agent, Network Mesh, Server IP, Server Hostname, Network Grid, autonomous tests (ASN, CDN, country, region, city), DNS Server Monitor, DNS Server Grid, HTTP(S) or API, Page Load and Transaction.

Estimated monthly test credits

Test	Interval	Per run	Credits / month
Mesh between 2 private agents	every 1 min	2 x 1 subtests x 2.5 = 5 per run	216,000
HTTP(S) tests to 2 SaaS app(s)	every 5 min	2 private + 3 global agents, 5 s timeout = 40 per run	345,600
BGP Monitor (1 test)	continuous	0.53 credits per minute	22,896
Total			584,496

Credit rules from the KB: network subtest 2.5 credits from a private agent and 5 from a global agent; HTTP subtests cost timeout (seconds) x 0.5 private or x 1 global; BGP Monitor 0.53 credits per minute; a test every minute runs 43,200 times in a 30-day month; Test Preview is free until saved. Mesh estimates count each direction as one subtest (the KB example: a 4-agent private mesh is 30 credits per run). Treat these as planning estimates and confirm with Kentik.

Source: Kentik KB, Synthetics Overview; Test Types Reference; Test Credits; Agents Page

10. Go-live validation checklist

☐	Branch 1: devices show Ashburn DC as their collection agent; flow, traps and syslog arrive across the WAN
☐	Branch 3: devices show Ashburn DC as their collection agent; flow, traps and syslog arrive across the WAN
☐	Branch 4: devices show Ashburn DC as their collection agent; flow, traps and syslog arrive across the WAN
☐	Ashburn DC: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Ashburn DC: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Ashburn DC: devices discovered in NMS with interfaces and health metrics
☐	Ashburn DC: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	Ashburn DC: routers peered with the agent (BGP Proxy); flows show AS path
☐	Data center 2: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Data center 2: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Data center 2: devices discovered in NMS with interfaces and health metrics
☐	Data center 2: test trap or syslog message visible (SNMP Traps: Events, Syslog: Events)
☐	Data center 2: BGP sessions established with Kentik; flows show AS path
☐	Branch 2: agent shows Up in Settings > Universal Agents, capabilities show enabled
☐	Branch 2: flow arrives in Data Explorer; device sample rates match Kentik; flow within about 20% of SNMP interface traffic
☐	Branch 2: devices discovered in NMS with interfaces and health metrics
☐	AWS: cloud export status OK; metadata visible in Kentik Map; flow logs arriving
☐	Azure: cloud export status OK; metadata visible in Kentik Map; flow logs arriving
☐	Synthetics: tests show Healthy, Warning or Critical (not Pending); agents Active
☐	Alerting: at least one alert policy routed to the customer's notification channel

Source: Kentik KB, UA Portal Management; Flow Overview; SNMP Traps & Syslog; Public Clouds

11. Security, data handling and licensing

Area	Detail
Hosting	US: Ashburn, VA. EU: Frankfurt. Chosen when the contract is signed.
In transit	TLS 1.2+ to and from Kentik
At rest	AES-256, consistent with FIPS 140-2
Isolation	Separate logical database and tables per customer
Audits	SOC 2 Type II yearly, annual third-party penetration test
Access	SAML 2.0 SSO, MFA (TOTP or hardware keys), user, admin and super admin roles, per-user API tokens
Data	IP addresses and network metadata, never packet payloads; IP redaction available
Retention (default)	45 days full-resolution flow, 120 days summarized, 3 or 13 months NMS metrics by plan; contracts can change this

Source: Kentik KB, Security Overview; KDE Overview

Licensing

License	Covers
FlowPak	Flow from routers, switches, firewalls and hosts, including SNMP flow enrichment.
NMS / DevicePak	Full device monitoring, plus syslog, SNMP traps and SSH config and commands.
CloudPak	Public cloud flow logs, sized in flows per second.
Synthetics	Monthly test credits: global agent subtest 5 credits, private agent 2.5, BGP Monitor 0.53 per minute.

Plans and editions vary by contract. Confirm entitlements with the Kentik account team.

Source: Kentik KB, Licenses; Test Credits

12. Open items to confirm with Kentik

- Which continuous synthetic test types run on the Universal Agent today (the KB names the capability Connectivity Test).
- Licensing names and entitlements per contract (FlowPak, NMS/DevicePak, CloudPak, Synthetics credits).
- Disk sizing for Universal Agent hosts is not published in the KB.
- BGP Proxy shows in Installed Capabilities in the portal, but the KB page still lists it as coming soon. Confirm it is available for this account before relying on it.
- Flow Proxy: Forwarding is marked beta in the KB. Confirm the flavor (Forwarding or Edge Processing) per site.

13. Appendix: Kentik KB references

Page	URL
Universal Agents	https://kb.kentik.com/docs/universal-agents
Public Clouds	https://kb.kentik.com/docs/public-clouds
Synthetics Overview	https://kb.kentik.com/docs/synthetics-overview
UA Deployment	https://kb.kentik.com/docs/ua-deployment
NMS Setup	https://kb.kentik.com/docs/nms-setup
SSH Command Access for Kentik NMS	https://kb.kentik.com/docs/ssh-command-access-for-kentik-nms
Kentik for AWS	https://kb.kentik.com/docs/kentik-for-aws
Kentik for Azure	https://kb.kentik.com/docs/kentik-for-azure
Test Types Reference	https://kb.kentik.com/docs/test-types
Azure Prerequisites & Roles	https://kb.kentik.com/docs/azure-prerequisites-roles
UA Portal Management	https://kb.kentik.com/docs/ua-portal-management
Device Settings	https://kb.kentik.com/docs/device-settings
Router Configuration	https://kb.kentik.com/docs/router-configuration
Flow Overview	https://kb.kentik.com/docs/flow-overview
Migration Scenarios (kproxy to UA)	https://kb.kentik.com/docs/kproxy-migration-scenarios
SNMP Traps & Syslog	https://kb.kentik.com/docs/snmp-traps-syslogs
Test Credits	https://kb.kentik.com/docs/test-credits
Metadata Configuration (AWS)	https://kb.kentik.com/docs/metadata-configuration-aws
Flow/Firewall Log Collection Configuration (AWS)	https://kb.kentik.com/docs/flow-firewall-log-collection-configuration
Kentik Export Configuration for AWS	https://kb.kentik.com/docs/kentik-export-configuration-aws
Flow/Firewall Log Collection (Azure)	https://kb.kentik.com/docs/flow-firewall-log-collection
Kentik Export Configuration for Azure	https://kb.kentik.com/docs/kentik-export-configuration-azure
Automated Configuration (PowerShell)	https://kb.kentik.com/docs/automated-configuration-powershell
Agents Page	https://kb.kentik.com/docs/agents-page
Security Overview	https://kb.kentik.com/docs/security-overview
KDE Overview	https://kb.kentik.com/docs/kde-overview
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